Artificial Intelligence in Forensic Medicine and Toxicology

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Introduction

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems capable of performing tasks that typically require human cognitive abilities, such as learning, reasoning, problem-solving, perception, and interaction. AI is not a single technology but a collection of methodologies, algorithms, and tools working together to achieve intelligent behavior.

Significant key components and characteristics of AI include:

- Machine Learning (ML): Enables systems to learn from data and improve over time without explicit programming.
- Neural Networks and Deep Learning: Processes complex data like images, audio, or text.
- Natural Language Processing (NLP): Enables machines to understand, interpret, and generate human language.
- Computer Vision: Interprets visual information from the world, such as images or videos.
- Robotics: Combines perception, decision-making, and action to interact with the environment.
- Reasoning and Decision-Making: Enables systems to make decisions based on predefined rules or learned patterns.
- Adaptability: Refines performance as the system processes more data.

AI is a rapidly evolving subdiscipline of computer science that has revolutionized multiple fields—and forensic medicine is no exception. In this domain, AI is utilized across three primary areas: teaching, research, and medicolegal work. This review article examines the role of AI in Forensic Medicine and Toxicology (FMT), highlighting its applications, ethical implications and limitations in medicolegal work.

Applications

AI enhances the accuracy, efficiency and objectivity of forensic investigations. In forensic medicine, AI systems—particularly those based on machine learning and deep learning—are capable of analysing large volumes of complex data far more rapidly than traditional methods. This supports forensic experts in tasks involving X-rays or images, such as fingerprint identification, bitemark analysis, age estimation based on radiology and injury interpretation. Feras Alafer, in a systematic review, concluded that while AI can influence diagnostic accuracy and efficiency in forensic medicine, it still requires enhanced standardization and comprehensive validation. [1]

Similarly, deep learning models have been successfully applied to classify gunshot wounds, distinguishing entry and exit wounds and estimating firing distance with high accuracy.[2] Toxicology is another area where AI has significant medicolegal utility; by analysing the patients clinical features, examination findings and reports of biological samples such as blood, urine, or hair, AI systems can rapidly detect drugs, poisons, and metabolites. This aids in determining the cause and manner of death and identifying poisoning cases with greater accuracy than conventional techniques.[3]

Furthermore, AI facilitates crime scene reconstruction by generating 3D models or animated simulations, allowing investigators and courts to better visualize events and understand the sequence of actions leading to a crime. [4] Another vital application is forensic radiology and postmortem imaging. Technologies such as CT and MRI, combined with AI algorithms, enable non-invasive "virtual autopsies," improving documentation while addressing cultural or

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religious concerns associated with traditional autopsies. AI also supports age estimation, sex determination and postmortem interval assessment by analysing skeletal and soft tissue features. Additionally, AI-powered microscopy can detect subtle evidence, such as blood spatter patterns or spermatozoa in sexual assault cases, that may not be visible to the human eye. [3]

Ethical implications

Development and validation of AI tools in medicolegal work is governed by strict ethical and regulatory frameworks. Permitted applications include:

- Language and Editorial Refinement: Using tools for grammar correction and stylistic improvements.
- Literature Synthesis: Summarizing background literature and assisting in figure generation under strict human oversight.
- Authorship Restrictions: In accordance with ICMJE guidelines, AI cannot be listed as an author, as authorship requires legal accountability and moral responsibility. [5]
- Transparency and Disclosure: AI-generated content must be explicitly disclosed in the methods or acknowledgments.
- Ethical Safeguards: Authors must prevent AI-driven plagiarism or "hallucinated" references. Compliance with data protection laws, such as the DPDP Act, is mandatory when sharing patient data.

Challenges and Limitations

Despite these advantages, AI integration faces challenges. A major limitation is the dependence on high-quality, well-annotated datasets; biased data can lead to erroneous conclusions with serious legal implications. Moreover, AI lacks the contextual understanding, ethical judgment, and empathy essential in medicolegal decision-making. Importantly, AI-generated findings are generally considered corroborative rather than definitive evidence in courts of law, as legal systems still rely on expert human testimony. [3]

Conclusion

AI has immense potential to augment medicolegal work by improving analytical capabilities and reducing workloads across various domains. However, its role should be viewed as complementary to, rather than a replacement for, human expertise. Careful integration and adherence to ethical standards are essential to ensure AI is used effectively in practice.

"AI won't replace humans, but humans who use AI will replace those who don't." — Sam Altman

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